7. SLIDING WINDOW

Aim:-

To write a program to implement flow control at data link y layer using SLIDING WINDOW PROTOCOL. Simulate the flow of frames from one node to another.

program Should achieve at least below give requirements. you can make it a bidirectional program wherein receiver is Sending its data frames with acknowlegement (Piggybacking).

Create a Sender program with following features:

1). Input window size from the user.

2) Input a Text message from the user.

3) Consider 1 character per frame

4) Create a frame with following fields [frame no. Data].

5) Send the frames. [print the output on Screen and Save it in a file called Sender-Buffer].

6) wait for the acknowlegement from the Receive.

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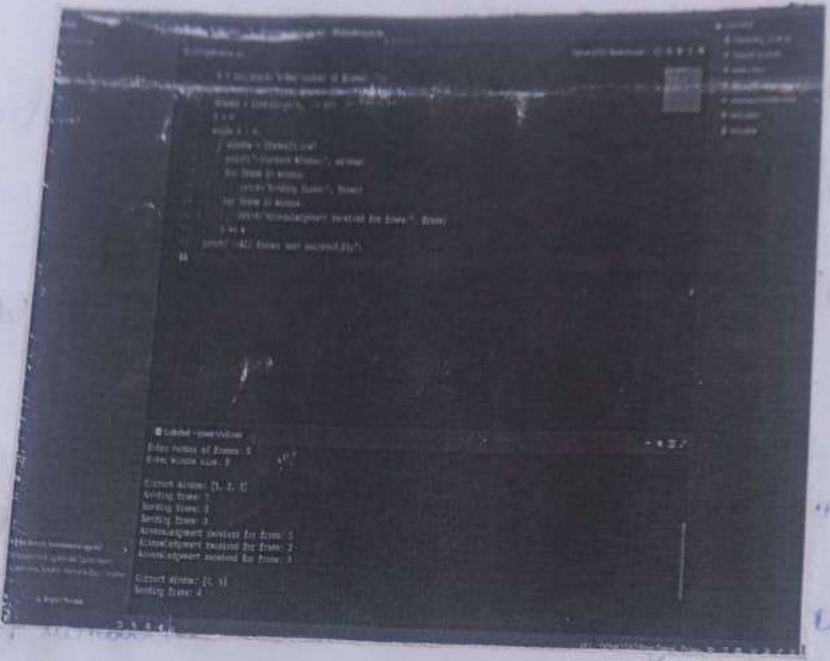

```
Enter number of frames: 5
Enter window size: 3

Current window: [1, 2, 3]
Sending frame: 1
Sending frame: 2
Sending frame: 3
Acknowledgement received for frame: 1
Acknowledgement received for frame: 2
Acknowledgement received for frame: 3

Current window: [4, 5]
Sending frame: 4
```

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7) Reader a file called Reciever buffer.

8) Check ACK feild the Acknowlegement number

9) If the Acknowlegement number is as expected Send new Set of frames accordingly.

Create a reciever file with following features:

1) Reader a file called Sender-Buffer

2) check the Frame no.

3) If the Frame no are as expected, write the appropriate ACK no in the Reciever-Buffer file. Else write NACK no. in the Reciever-Buffer file.

Sender code:

```
import os.
import time
Base_DIR = os.path.dirname (os.path.abspath ( )
printf ("Sender Buffer", Sender_File).
print ("Reciever Buffer", Reciever-File).
window_size = int (input ("Enter window size: ")).
message = input ("Enter the text message: ").
```

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```
if window-size <= 0;
    print("window size must be greather than zero.")
    false systemed;
frames [].
while Start < total-frames:
    end = minstart + window-size, total-frames).
    print("\n= = = = = = = = = = =   = = = =")
    print("Sending ~~names~~ frames", Start + 1, "to", end)
    print ("= = = = = = = = = = = = = = =")
  elif ack-type == "NACK"!
      print("NACK received")
      print( "Retransmelling from frame :", ack_no)
  else:
     print("unknown exponse: Retrying current window:")
print("\n . . . . . ")
print( f"Total frames", total-frames):
```

Reciever code.

```
import os.
import time.
expected-frame = 1
last.contein = "'
for link in lines
  try! parts = line . split(";",1)
  except (linkError, valueerror.
  break;
```

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```
else:
    print("Frame error detected")
    print("Expected frame:", expected_frame)
    error_found = True
print("\waiting for next window...")
time.sleep(0.5)
```

~~Input~~ .

Sender code Input

```
window size: 3
Message: HELLO
```

Sender code output

```
Frame 0: H
Frame 1: E
Frame 2: E
Frame 3: L
Frame 4: O
```

Receiver code Input.

Sender Buffer.txt

```
0  H
1  E
2  L
3  L
4  O
```

Receiver code output

```
Received 0: H
Received 1: E
Received 2: L
Received 3: L
Received 4: O
```

Result:

Therefore, the implement flow control of datalink Layer using Sliding window protocol executed successfully.

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